SGW Resilience

Product Requirements Document
AECOM AI Solution Engineer Case Assignment

1. Problem Definition & Business Context

Business context

SGW operates critical electricity and water infrastructure serving over eight million residents. Severe weather and rising operating pressures increase the need for effective preparedness and response.

Information is fragmented across GIS, maintenance, weather and field systems, limiting asset review, response coordination and shared situational awareness.

Product objective

Support flood preparedness and pump condition review through a shared decision-support workflow that enables teams to:

- Prioritize weather-exposed assets and review pump condition alerts with supporting asset context.
- Review recommended actions, assign approved work and track ownership and outcomes.
- Give leadership a timestamped overview of active issues and unresolved work.

2. Key Assumptions & Unknowns

The initial design relies on the following assumptions, which require confirmation with SGW before an operational pilot.

Assumption	What needs confirmation	Design implication
Data readiness: GIS, weather, maintenance and work data can be linked and updated.	Source access, data quality and acceptable update delays.	Agree integration and refresh rules before operational use.
Risk prioritization: Review criteria can be agreed with SGW.	Accepted factors and thresholds.	Treat prototype scoring as provisional until approved.
Pump telemetry: Suitable telemetry history is available; not guaranteed by the brief.	Availability and representative operating conditions.	Include condition monitoring only when a baseline can be validated.
Approved procedures: SGW can provide approved operating procedures.	Document availability, approval and ownership.	Ground recommendations in approved procedures.
Operational ownership: SGW can assign reviewers and task owners.	Approval responsibilities and team availability.	Align approval and task assignment with agreed roles.
External AI approval: SGW permits external AI for selected data.	Data handling restrictions and approved deployment options.	Confirm the approach before sending client data.

3. Target Users & Pain Points

The proposed users and needs below should be confirmed during client discovery.

Target user	Main pain point	What they need
Operations / Emergency Response Lead	Fragmented information makes response priorities difficult to establish.	Review asset priorities and supporting evidence, then approve response actions.
Maintenance / Condition Engineer	Unusual telemetry is difficult to assess without maintenance and operating context.	Investigate condition alerts using telemetry, asset history and relevant procedures.
Field Operations Supervisor	Scattered assignments and updates create unclear ownership and duplicate work.	Coordinate approved tasks, track deadlines and record progress and outcomes.
Leadership	Inconsistent updates obscure current priorities and unresolved issues.	A timestamped overview of active risks, response progress and outstanding decisions.

4. Functional & Non-Functional Requirements

Functional requirements

ID	Requirement	Acceptance criterion
FR-01	Identify and prioritize assets affected by a weather warning.	Display affected assets, priority, supporting reasons and data quality flags.
FR-02	Combine asset, maintenance and existing work information.	Users can inspect these records for the selected asset through a shared identifier.
FR-03	Detect persistent unusual pump behavior.	Evaluate valid telemetry windows against the model threshold and create an alert when the configured persistence rule is met.
FR-04	Produce evidence-supported action drafts.	Show proposed actions, source excerpts and missing information. Reject citation IDs outside the retrieved evidence and actions outside the approved catalog.
FR-05	Require human approval before creating response work.	Only selected actions become tasks. Reject approval when the alert or related signals have changed since drafting.
FR-06	Track response work through completion.	Tasks have an owner, deadline and status. Completion requires an outcome. Reuse an unfinished task for the same asset and action code.
FR-07	Support alert review and closure.	Users can acknowledge or close alerts. Closure is blocked while linked work remains incomplete.
FR-08	Provide a leadership overview and decision history.	Show active alerts, outstanding work and a timestamped briefing, with access to recorded decisions and actions.

The prototype uses three consecutive threshold exceedances. Operational thresholds and persistence rules require engineering validation. Source integrations are described in Section 6.

Non-functional requirements

ID / Area	Requirement
NFR-01 Performance	Agree response-time and concurrent-user targets with SGW before the pilot. Measure API, retrieval and generation latency against these targets under representative workloads.
NFR-02 Persistence and recovery	Preserve saved operational records across restarts. Agree availability and recovery targets with SGW before the pilot; verify backup and restore against the agreed recovery targets.
NFR-03 Data consistency	Failed operations must roll back. Repeated warning ingestion or plan approval must not create duplicate records.
NFR-04 Security	Protected API routes require authentication. Credentials remain server-side and must not appear in the UI or application logs.
NFR-05 Traceability	Record significant workflow actions with actor, timestamp and affected record. Retain the evidence and context used for each draft.
NFR-06 Validation and failure handling	Reject invalid inputs and return clear errors. Retrieval or generation failure must not create an action plan or tasks.

Access controls and prototype security limitations are described in Section 8.

5. Proposed AI Capabilities

Weather and condition alerts use the same asset context, human approval and response process. Condition alerts can occur without a weather warning and do not establish weather-related causation.

Capability	Expected behavior	Decision support
Pump anomaly detection (non-LLM)	Compare telemetry with a normal operating baseline. Flag persistent deviations and show scores, thresholds and supporting measurements.	Help engineers prioritize investigation. Does not diagnose faults or forecast failure.
Action drafting (LLM with retrieval)	Use approved procedures and operational context to draft actions with citations and missing information.	Help operators assess proposed actions and their supporting evidence.
Leadership briefing (LLM)	Summarize active alerts, outstanding work, ownership and uncertainty from a timestamped snapshot.	Support consistent situational updates for leadership.

Flood prioritization uses explicit rules, not AI forecasting. Pump condition alerts support a separate engineering review and do not change the flood priority index. AI capabilities require validation on representative SGW data before operational use.

6. High-Level Architecture & Integrations

The prototype runs locally in Docker Compose. A Streamlit interface calls a single FastAPI application containing the weather, condition and decision workflows. PostgreSQL stores operational data and workflow history.

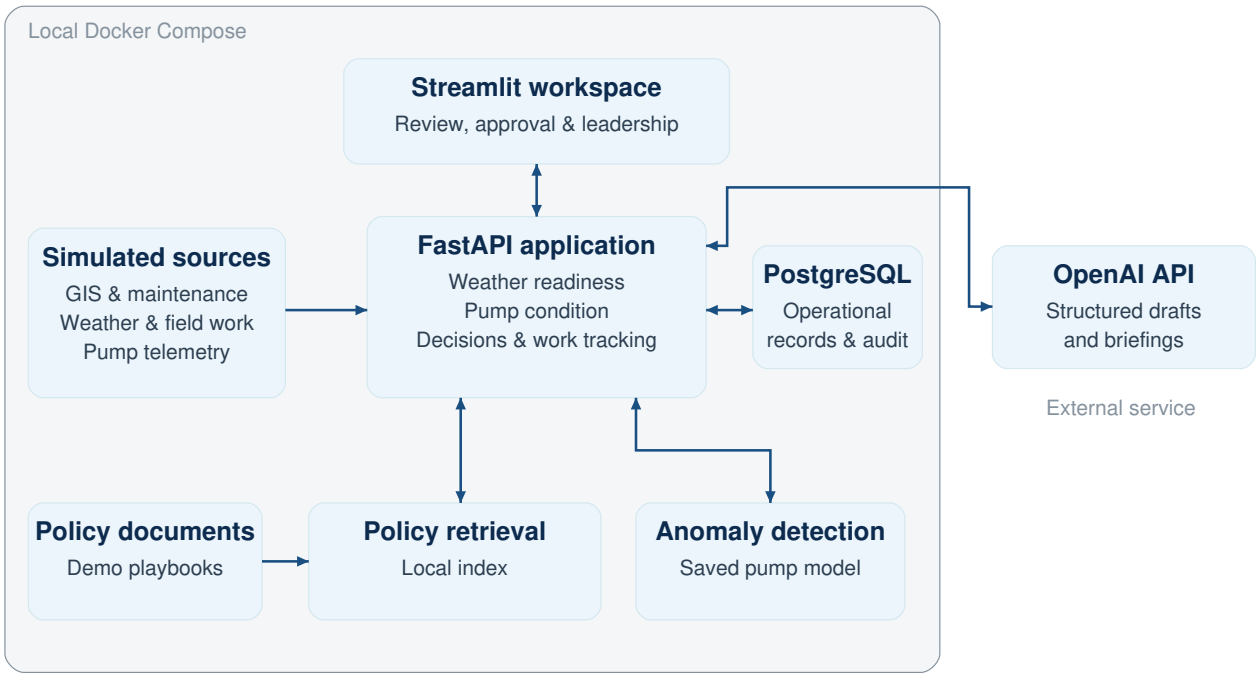


Figure 1: Local prototype architecture. Source inputs are simulated; the OpenAI connection is live.

Policy retrieval and anomaly inference run within the FastAPI backend. OpenAI is an external service used for structured action drafts and leadership briefings. Operational sources are simulated; source loading and workspace refresh are manual.

Integrations

Data source	Current prototype	Required pilot integration
GIS and maintenance	Synthetic records linked by asset ID.	Client system access, identifier mapping and agreed updates.
Weather	Simulated warning revisions.	Provider feed with geographic coverage, severity, validity and revisions.
Pump telemetry	Synthetic telemetry samples.	Suitable sensor connection and validated operating baseline.
Field operations	Seeded assignments and locally managed tasks.	Work-system integration and agreed synchronization rules.
Operating procedures	Versioned synthetic playbooks.	Access to approved documents and an update process.

7. Data Requirements & Dependencies

Asset-related records require consistent asset identifiers. Weather warnings are matched to asset locations using their geographic coverage.

Data category	Required information
Asset register / GIS	Asset ID, type, location, business criticality, hazard exposure, backup availability and last update time.
Maintenance records	Record ID, asset ID, issue description, priority, status and last update time.
Weather warnings	Event ID, revision, hazard type, severity, affected area, source and validity times.
Pump telemetry	Asset ID, timestamp, running state, speed, current, flow and motor temperature.
Field work	Task ID, asset ID, action, owner, deadline, status and completion outcome.
Operating procedures	Approved document content, document ID, version, applicable hazard and asset type.

Data quality

- Map asset identifiers consistently across systems.
- Use consistent timestamps and documented measurement units.
- Flag missing, invalid or stale data against freshness limits agreed with SGW.

Telemetry sampling frequency, evaluation windows and supported operating conditions will be defined with SGW engineers and validated against available equipment data.

Dependencies and readiness

Named data owners must confirm source availability and quality. Pump monitoring requires representative historical telemetry and engineer-reviewed examples of normal and unusual operation to establish baselines and evaluate detection quality.

8. Security, Governance & Human-in-the-Loop

The platform provides decision support, with no direct equipment control or automatic field dispatch.

Area	Required controls
Access control	Use verified identities and role-based permissions. Agree approval responsibilities with SGW.
Data protection	Keep credentials server-side and minimize data shared with AI services. Agree permitted data use, access, retention and hosting with SGW.
Human approval	Require explicit approval of selected actions after reviewing evidence and missing information. Block approval of outdated drafts and work based on expired warnings.
Traceability	Retain draft context, source excerpts and document and alert versions. Record approvals, task updates and alert disposition with actor, timestamp and affected record.
AI oversight	Use approved procedures and a restricted action catalog. Monitor output quality and test model, prompt and procedure changes, including unsupported recommendations and prompt injection. Assign oversight and maintenance owners.

Prototype controls and limitations

The prototype uses a shared API token and manually entered reviewer names. Verified identities, role-based permissions and protection against audit alteration require further implementation.

Citation and action-catalog checks constrain generated outputs but do not establish correctness. Reviewers must confirm that the evidence supports each recommendation.

SGW security, legal and operations teams must confirm applicable regulatory and contractual requirements and approve the controls before operational data is used.

9. Success Metrics, MVP Scope & Delivery Priorities

Success metrics

Success will be assessed through a controlled pilot using representative data and agreed operational scenarios.

Outcome	Proposed measure
Faster prioritization	Elapsed time from warning receipt to an engineer-reviewed asset list, compared with the current process.
Useful action drafts	Share of reviewed drafts accepted, corrections per draft and share of reviewed recommendations supported by cited procedures.
Response coordination	Share of tasks due in the measurement period completed on time; duplicate assignments in that period.
Condition alert quality	Share of engineer-reviewed anomalous test cases detected and false alerts during confirmed normal operation. Measurement periods will be agreed with SGW.
Reliable leadership updates	Preparation and review time; factual accuracy, completeness and visible data freshness.

If pump monitoring is included in the pilot, evaluate detection on held-out data within supported operating conditions. Count each alert episode once; repeated updates are not new alerts.

SGW operations and engineering leads will agree baselines, test sets, measurement periods and acceptance targets before the pilot. Synthetic checks demonstrate workflow behavior, not client outcomes or equipment-model accuracy.

MVP scope

The proposed MVP supports flood readiness in one operating region for selected pumping stations and substations. It combines asset, warning, maintenance and existing work information in a shared response workflow.

Direct equipment control, automatic field dispatch, weather forecasting, failure prediction and organization-wide rollout are outside the initial scope.

Delivery priorities

Priority	Scope
First: Core flood-response workflow	Deliver warning ingestion, asset review, evidence-supported drafts, human approval, task tracking and leadership updates for the selected asset set.
Conditional: Pump monitoring	Include for an agreed pump family and operating conditions once telemetry readiness and model quality meet agreed acceptance criteria.
Later: Expansion	Extend to additional regions, hazards, asset types and AI capabilities, based on pilot results, data readiness and business value.

Operational use requires agreed data access, operating rules and security controls, as defined in Sections 2, 7 and 8.